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Improved anaerobic digestion performance of *Miscanthus floridulus* by different pretreatment methods and preliminary economic analysis

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ABSTRACT

During the anaerobic digestion (AD) of cellulosic substrate, an economical and environmental friendly pretreatment method should be selected to improve the AD performance. In this study, five pretreatment methods include four traditional pretreatment methods (alkali hydrogen peroxide pretreatment, NaOH pretreatment, HCl pretreatment and hot water pretreatment) and one new developed pretreatment (microaerobic pretreatment) method were selected to study the effects of pretreatment on the AD of *Miscanthus floridulus*. In addition, preliminary economic analysis of different pretreatment methods was also conducted. Results showed pretreatment could significantly improve the AD performance of *Miscanthus*. After alkali hydrogen peroxide pretreatment, microaerobic pretreatment, NaOH pretreatment, HCl pretreatment and hot water pretreatment the methane yields of *Miscanthus* were 41.1%, 25.5%, 18.7%, 15.7% and 10.2% higher than that of untreated sample. Though all these five pretreatment methods could improve the methane yield significantly, their action sites were different. H_2O_2 pretreatment and NaOH pretreatment were efficient in removal lignin from *Miscanthus floridulus*. Hemicellulose structure was also partly destroyed during H_2O_2 pretreatment and NaOH pretreatment. HCl pretreatment was efficient in destruction of hemicelluloses structure and partly destroyed the cellulose structure. Microaerobic pretreatment and hot water pretreatment were mainly acted in partly destruction of hemicelluloses and cellulose components. Preliminary economic analysis showed compared with un-pretreated group only microaerobic pretreatment was economically feasible. In addition, compared with other pretreatment methods, microaerobic pretreatment just needs limited amount of oxygen, which is more environmental friendly.

1. Introduction

Due to the energy shortage and environmental pollution, bio-methane production from organic waste through AD method has attracted increasing attention [1–3]. *Miscanthus* is a tall perennial grass with high water and nitrogen use efficiency [4–6]. In addition, biomass yields of *Miscanthus* is much higher, *Miscanthus* can also well adapt to different climate and soil conditions [4,6]. These all make *Miscanthus* to be a potential substrate for biogas production. *Miscanthus* is a lignin rich substrate, and the hemicellulose in *Miscanthus* was encircled by compact lignin structure. Therefore, the AD of *Miscanthus* is always inefficient [7]. During the AD of *Miscanthus*, appropriate pretreatment is essential to destroy the compact structure and improve the AD efficiency [5].

Using pretreatment methods to improve the AD performance of cellulosic substrate has been widely studied [8–10]. Alkali pretreatment, acid pretreatment, alkali hydrogen peroxide pretreatment, hot water pretreatment and biological pretreatment are among the most studied pretreatment method. Alkali pretreatment mainly acts in removal/destruction the lignin structure or dissolution the hemicelluloses structure [11,12]. The most common alkalis during alkali pretreatment were NaOH, KOH, $Ca(OH)_2$ and $NH_3{\cdot}H_2O$, among which NaOH was the most studied and was thought to be the most efficient [13]. Acid pretreatment has little impact on the lignin structure of cellulosic substrate, which mainly acted with the cellulose structure. After acid pretreatment, the cellulose structure was partly destroyed or became more sensitive to relative enzyme [13]. Both inorganic (e.g. H_2SO_4, HCl, HNO_3, H_3PO_4) and organic acids (e.g. acetic acid, maleic acid) were widely used during acid pretreatment. Alkali hydrogen peroxide pretreatment could destroy the lignin structure of cellulosic substrate by the strong oxidizing property of hydrogen peroxide. During hydrogen peroxide pretreatment, the pH of pretreatment system was a

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critical factor, studies showed at the pH of 11.8 the delignification was best [14]. During hot water pretreatment, water could penetrate into the cellulosic structure and destroy hemiacetal linkages of hemicelluloses, which could liberate O-acetyl and other acid moieties. The released acids in turn could accelerate the removal of oligosaccharides from hemicellulose [15]. Biological pretreatment mainly achieved by addition of bacterial or fungal strain or a consortium of bacteria and/or fungi, with high capacity of degrading cellulose/hemicelluloses/lignin [16]. An advanced biological pretreatment using fungal solid state fermentation was reported by Yahmed et al., which enhanced biogas production of 21% [17]. There were no toxic compound productions and additional chemical needed during biological pretreatment and the biological pretreatment could conduct at mild condition. Therefore, biological pretreatment received increasing attentions.

Capital cost and environmental implication of the applied pretreatment method are also a critical aspect except the improved AD efficiency. During the application of conventional pretreatments method, additional energy or chemicals are required, which may lead to increased capital cost. The study of Passos et al. [18] showed although thermochemical pretreatment increased the methane potential of dairy cow manure, AD without pretreatment was more economical. In addition, inhibitory and biologically non-degradable compounds may also generate during these traditional pretreatment method. Michalsk and Ledakowicz [19] found although H_2O_2 decomposition could degrade energy plant significantly, however, the biogas production was strongly inhibited by byproducts of the pretreatment process. These all hold up the engineering application of these pretreatment methods. Therefore, an economic feasible and environmental friendly pretreatment method still needed to develop. Recently studies showed a novel biological pretreatment-microaerobic pretreatment is also an alternative pretreatment method both for solid substrate (e.g. cellulosic substrate) and liquid substrate (e.g. food waste and brown water) during the AD process [20–22]. During microaerobic pretreatment, aerobic/facultative microorganisms could be enriched and keep high activity, which would in turn produce more extracellular enzymes (e.g. cellulase). The produced extracellular enzymes could react with substrate and therefore improve the hydrolysis rate. As a novel pretreatment, microaerobic pretreatment has several advantages e.g. no additional chemicals need, moderate environmental condition. Therefore, further study of microaerobic pretreatment is needed.

In the present study, in order to select an economically feasible and environmental friendly pretreatment method for the AD of *Miscanthus*, the technical and preliminary economical evaluation of five pretreatment methods include four traditional pretreatment methods (alkali hydrogen peroxide pretreatment, NaOH pretreatment, HCl pretreatment and hot water pretreatment) and one new developed pretreatment (microaerobic pretreatment) method were conducted. The effects of different pretreatment on the structure, methane yield and fermentation kinetics of *Miscanthus* were studied and compared.

2. Methods and materials

2.1. Substrate and inoculum during AD

Miscanthus floridulus used in this study was collected from HuNan province (China), before further use, *Miscanthus floridulus* was air dried to a TS (Total Solid) of 83.45 ± 0.6% with VS of 93.70 ± 0.64%. The hemicelluloses, cellulose and lignin content of *Miscanthus floridulus* were 23.78 ± 0.19%, 30.15 ± 0.05% and 25.91 ± 0.09% (based on TS), respectively.

Inoculum used in this study was biogas slurry, collected from a 500 m^3 size of biogas plant (Qingdao, Shandong province, China). The biogas plant used corn straw as feedstock and running at 35 °C with Hydraulic Retention Time (HRT) of 35d. The TS and VS of biogas slurry were 6.72 ± 0.1% and 78.73 ± 0.1% (based on TS), respectively. The collected biogas slurry was stored in a refrigerator at 4 °C, before AD, the biogas slurry was pre-incubated at 37 °C for 3 days.

2.2. Pretreatment of Miscanthus floridulus

In this study, five pretreatment methods were selected: NaOH pretreatment, HCl pretreatment, hot water (low temperature) pretreatment, alkali H_2O_2 pretreatment and thermophilic microaerobic pretreatment. During NaOH pretreatment, 6% NaOH solution was used, the liquid to solid ratio was 10 ml/g and performed at 35 °C for 3 h [23]. During HCl pretreatment, the pH of pretreatment system was adjusted to 1 with HCl, liquid to solid ratio was 10 ml/g and performed at 99 °C for 30 min [24]. The hot water (low temperature) pretreatment was performed at 95 °C for 10 h, the liquid to solid ratio was 10 ml/g [25,26]. During alkali H_2O_2 pretreatment, 2% H_2O_2 solution was used, the liquid to solid ratio was 10 ml/g and performed at 35 °C for 24 h, before pretreatment the pH of this pretreatment system was adjusted to 11.8 with 2 M NaOH solution [27]. Microaerobic pretreatment was conducted in 300 ml digester bottles, during which biogas slurry was used as inoculum (the TS ratio of inoculum and substrate was 1:4) and the oxygen load of microaerobic pretreatment was set as 5 ml/g VS (VS of substrate). The oxygen needed was injected into the digester bottles by syringe and then the digester bottles were placed in a shaking water bath at 55 °C [22].

2.3. Batch AD of Miscanthus floridulus

Quadruplicate AD of *Miscanthus floridulus* was conducted in 300 ml digester bottles with a working volume of 200 ml. After pretreatment, pH of the system was adjusted to 7.2 with 2 M NaOH and 2 M HCl. After that 40 ml biogas slurry was added, the working volume during AD was adjusted to 200 ml with nutrient solution as Angelidaki et al. [28] described. Batch AD of *Miscanthus floridulus* was conducted in shaking water bath at 37 °C with 120 rpm.

2.4. The structure analysis of Miscanthus floridulus

In this study, the structure changes of *Miscanthus floridulus* were analyzed by wide angle X-ray diffraction (XRD, Bruker D8 Advance) and Scanning electron microscopy (SEM, S-4800 Hitachi). The crystallinity index (CrI) of *Miscanthus floridulus* was calculated as following: $CrI = [(I_{002} - I_m)/I_{002}] \times 100$. Among which I_{002} is the intensity for the crystalline portion at about $2\theta = 22$ and I_m is the peak for the amorphous portion at about $2\theta = 16.2$ [29].

2.5. Equation used to analyze the methane production kinetics

In order to analyze the methane production kinetics of *Miscanthus floridulus*, the *Modified Gomperz equation* as Zhang et al. [30] and Cai et al. [31] described was used to fit the methane production data. The *Modified Gomperz equation* was written as following:

$$P(t) = P\exp\left[-\exp\left(\frac{Rm*e}{P}(\lambda - t) + 1\right)\right]$$

Among which $P(t)$, P, Rm, λ and t represent cumulative total methane production (ml/g $VS_{substrate}$), total methane production potential (ml/g $VS_{substrate}$), maximum methane production rate (ml/d/g $VS_{substrate}$), lag-phase time (d) and elapsed time (d), respectively.

2.6. Preliminary economic analysis of different pretreatment methods

Preliminary economic analysis in this study was based on data obtained from batch experiments and model fitting results. According to a previous study, the hydraulic retention time (HRT) was assumed to be the ratio between P and Rm (fitted results of *Modified Gomperz equation*) [18]. During the Preliminary economic analysis process, the AD system

was assumed to run under the ideal condition, which means there were no energy loses during the pretreatment/start-up/running stage.

The energy needed (Q) for heating was calculated as following:

$$Q = \rho * V * C * (T_{final} - T_{initial})$$

where Q is the energy needed for heating (kJ), ρ is the density of the system ($kg \cdot m^{-3}$), V is the volume of reactor (m^3), C is the specific heat of the system ($kJ/(kg * °C)$), T_{final} is the finial temperature of the system (°C), $T_{initial}$ is the initial temperature of the system (°C). In this study, ρ and C of the system was set as same with water, which is 1000 $kg \cdot m^{-3}$ and 4.2 $kJ/(kg * °C)$. During the pretreatment and start-up/running stages, the energy needed (Q) was provided by electric energy, the energy transfer efficiency was set as 80%. The electricity price in China is 0.09 U.S.$/kWh.

The Preliminary economic was based on the economic input during pretreatment, start-up/running stage of AD (include the economic input for chemical and the energy needed for heating) and methane yield, which was conducted as following:

$$E = \frac{Q1 + C + Q2}{Y}$$

where E (U.S.$/$m^3$) is the cost to produce per cubic meter biomethane, Q1 (U.S.$) is the cost for heating during pretreatment stage, C (U.S.$) is the cost for chemical during pretreatment stage, Q2 (U.S.$) is the cost for heating during the start-up and running stage of AD, Y (m^3) is the cumulative methane yield during the AD process.

During the running stage of AD the energy needed for mixing/feed was not counted in. Assumptions for the Preliminary economic analysis were shown in Table 1.

2.7. Analytical methods

During this study, biogas yield was tested by water replacement method. The detail experimental apparatuses of water replacement method were shown in Fig. S1. A gas chromatograph (SP 6890, Shandong Lunan Inc., China) with argon as carrier gas was used to test the methane concentration in biogas. The used gas chromatograph equipped with a Porapak Q stainless steel column (180 cm long, 3 mm outer diameter) and TCD (thermal conductivity detector). As for the injector, detector, and oven the temperatures were 50, 100 and 100 °C, respectively. For TS test the material was dried at 105 °C for 8 h, for VS test the dried material was ignited at 550 °C for 2 h. The method as Goering and Van Soest [32] described was used to test the hemicelluloses, cellulose and lignin contents of *Miscanthus floridulus*. When testing SCOD (soluble chemical oxygen demand), samples were firstly centrifuging at 10,000 rpm for 5 min and filtering samples through a 0.45 μm glass microfiber filter and determined by spectrophotometric method [33]. The pH value was tested by a pH meter (B-711, Horiba, Japan).

Table 1
Assumptions for the preliminary economic analysis.

	Value and unit
Scale of reactor	500 m^3 with a working volume of 80%
TS of the AD system	4%
Substrate	*Miscanthus floridulus*
Inoculum	Biogas slurry
Running temperature	37 °C
Hydraulic retention time (HRT)	Ratio between *P* and *Rm* [16]
Energy transfer efficiency	80%
Cost for substrate (*Miscanthus floridulus*)	27.22 U.S.$/ton
Price of electricity	0.09 U.S.$/kWh
Price of NaOH	483.95 U.S.$/ton
Price of HCl	725.93 U.S.$/ton
Price of H_2O_2 (50%)	260.88 U.S.$/ton

Note: During the Preliminary economic analysis, pretreatment and start-up/running stages were assumed to proceed under ideal condition, which means there are no energy loses during these stages. The energy needed for mixing system, feed system during AD was not counted in. The initial temperature of system was set as 25 °C.

3. Results and discussion

3.1. The structure changes of Miscanthus floridulus after different pretreatment methods

3.1.1. The effect of pretreatment on the hemicelluloses, cellulose and lignin contents of Miscanthus floridulus

Cellulosic substrates like *Miscanthus floridulus* are mainly composed by hemicelluloses, cellulose and lignin, these three components hinge with each other by non-covalent and covalent cross-linkage, which makes the structure of cellulosic substrate stable [34,35]. Pretreatment methods are widely used to improve the AD efficiency mainly by destruction the strong interlinkage among these three components, breaking down lignin interference, reducing cellulose crystallinity or increasing surface area [36,37]. The hemicelluloses, cellulose and lignin contents of *Miscanthus floridulus* before and after pretreatment were shown in Table 2. "Degradation rate" in Table 2 means the percents of cellulose, hemicelluloses, lignin removed during pretreatment process. It was calculated as following (taking cellulose as example): Degradation rate = (initial percent of cellulose * initial TS of *Miscanthus* – percent of cellulose after pretreatment * TS of *Miscanthus* after pretreatment)/initial percent of cellulose * initial TS of *Miscanthus*.

Pretreatment showed obvious impact on the hemicelluloses, cellulose and lignin contents of *Miscanthus floridulus*. Among these 5 pretreatment methods, H_2O_2 pretreatment and NaOH pretreatment were efficient in removal lignin from *Miscanthus floridulus*. Also, the hemicellulose structure was partly destroyed during H_2O_2 pretreatment and NaOH pretreatment. HCl pretreatment was efficient in destruction of hemicelluloses structure and partly destroyed the hemicellulose structure. Microaerobic pretreatment and hot water pretreatment were mainly acted in partly destruction of hemicelluloses and cellulose components. However, the impact of hot water pretreatment on the hemicelluloses and cellulose contents of *Miscanthus floridulus* was less, which might due to the running condition of hot water pretreatment was mild (95 °C for 10 h) compared with typical hot water pretreatment (usually at higher temperature e.g. 150 °C [38], 190 °C [39]). Results obtained from this study were quite consistent with previous studies, NaOH pretreatment mainly acted in lignin removal from cellulosic substrate [12,40], H_2O_2 pretreatment acted efficiently in destruction of lignin structure[27,41], HCl pretreatment and microaerobic pretreatment mainly acted in destruction of hemicelluloses and cellulose structure [22,42]. The destroyed *Miscanthus floridulus* after pretreatment was more accessible for enzymes during the AD process (especially the hydrolysis process), which led to higher methane production rate and finally led to increased methane yield (shown as Figs. S2, 3 and Table 4).

3.1.2. The effect of pretreatment on the crystallinity index of Miscanthus floridulus

The crystallinity of the cellulosic substrate is broadly accepted to be unfavorable for the enzyme access [43], which is an important factor during the AD process. The crystallinity indexes of *Miscanthus floridulus* before and after pretreatment were shown in Table 3 and Fig. 1. Different pretreatment methods showed unequal impacts to the crystallinity index of *Miscanthus floridulus*. After NaOH pretreatment and H_2O_2 pretreatment, the crystallinity index of *Miscanthus floridulus* increased. Relatively, after hot water pretreatment, microaerobic pretreatment and HCl pretreatment the crystallinity index of *Miscanthus floridulus* decreased. Unequal impacts to the crystallinity index of different pretreatment methods correspond with the mechanisms of these pretreatment methods. Crystallinity index ($CrI = [(I_{002} - I_m)/I_{002}] \times 100$) of *Miscanthus floridulus* was affected by both the contents of amorphous

Table 2
The effect of pretreatment on the hemicelluloses, cellulose and lignin contents of *Miscanthus floridulus*.

Groups	Hemicelluloses (%, based on TS)	Cellulose (%, based on TS)	Lignin (%, based on TS)	Degradation rate (%, relative to control)		
				Hemicelluloses	Cellulose	Lignin
Control	23.78 ± 0.19	30.15 ± 0.05	25.91 ± 0.09	0.00	0.00	0.00
NaOH pretreatment	21.60 ± 0.73	31.63 ± 0.71	23.98 ± 0.41	13.43	0.01	11.79
H_2O_2 pretreatment	28.29 ± 1.16	43.12 ± 0.21	10.89 ± 1.29	16.97	0.18	70.67
Hot water pretreatment	22.29 ± 0.12	29.38 ± 0.17	25.56 ± 0.43	5.04	1.27	0.06
Microaerobic pretreatment	22.68 ± 0.34	28.14 ± 1.29	25.31 ± 0.58	5.16	7.19	2.87
HCl pretreatment	18.04 ± 1.46	28.89 ± 0.92	28.02 ± 0.33	30.26	11.91	0.58

Table 3
Crystallinity index of pretreated and untreated *Miscanthus floridulus*.

	I_{002}	I_m	CrI	Relative change (%, relative to control)
Control	4482	2693	39.9	0.0
NaOH pretreatment	4324	2549	41.1	3
H_2O_2 pretreatment	4519	2546	43.7	9.5
Hot water pretreatment	3732	2292	38.6	−3.3
Microaerobic pretreatment	4171	2662	36.2	−9.3
HCl pretreatment	3851	2445	36.5	−8.5

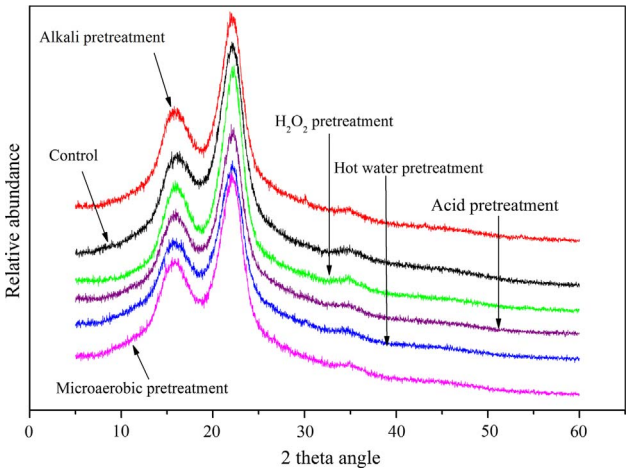


Fig. 1. XRD patterns of untreated and pretreated *Miscanthus floridulus*.

structure (e.g. amorphous cellulose, hemicellulose and lignin) and crystalline structure (crystalline cellulose) [44–46]. As can be seen from Table 2, NaOH pretreatment and H_2O_2 pretreatment mainly acted in lignin and hemicellulose removal or destruction of lignin structure from cellulosic substrate, they had little effect on crystalline cellulose structure. The removal of amorphous structure (lignin or hemicellulose) might be the reason for the increased crystallinity index after NaOH pretreatment and H_2O_2 pretreatment. Hot water pretreatment, microaerobic pretreatment and HCl pretreatment were efficient in destruction of cellulose structure. However, they had little effect on amorphous structure (e.g. lignin). Therefore, after hot water pretreatment, microaerobic pretreatment and HCl pretreatment the crystallinity index of *Miscanthus floridulus* all increased. The XRD analyses results were quite consistent with the structure analyses results (shown in Table 2). The partly destructions of compact structure during pretreatment process were the basis for the improved AD performance of pretreated *Miscanthus floridulus*.

3.1.3. The effect of pretreatment on the microstructure of Miscanthus floridulus

Pretreatment was also widely used to increase the surface area of cellulosic substrate, therefore improved the enzyme access and finally improved the AD efficiency. In this study, the microstructures of pretreated and untreated *Miscanthus floridulus* were observed by SEM, the SEM images were shown in Fig. 2. Before pretreatment, *Miscanthus floridulus* was intact with smooth and compact surface, there were even no pores on the surface of *Miscanthus floridulus*. After pretreatment, the intact surface of *Miscanthus floridulus* was destroyed. Especially after NaOH pretreatment and H_2O_2 pretreatment, many pores appeared on the surface, which might cause by the removal of lignin during these two pretreatment. The destroyed intact surface and appeared pores could greatly enhance the enzyme access. Therefore the AD of pretreated *Miscanthus floridulus* was more efficient, which also led to higher methane yield from pretreated *Miscanthus floridulus*.

3.2. Methane production from Miscanthus floridulus

The cumulative methane yields during AD of *Miscanthus floridulus* were shown in Fig. 3. Results showed pretreatment before AD led to improved methane yield from *Miscanthus floridulus*. Among these five pretreatment methods, methane yield of H_2O_2 pretreated *Miscanthus floridulus* was highest, which was 334 ml/g $VS_{substrate}$ and 41.1% higher than that of control (without pretreatment). Methane yield from NaOH pretreated *Miscanthus floridulus* was in second place, which was 297.1 ml/g $VS_{substrate}$ and 25.5% higher than that of control. Methane yields from microaerobic pretreated, HCl pretreated and hot water pretreated *Miscanthus floridulus* were 281, 273.8 and 260.9 ml/g $VS_{substrate}$ respectively, which were 18.7%, 15.7% and 10.2% higher than that of control, respectively. Statistical analysis of the methane yields from *Miscanthus floridulus* with different pretreatments were conducted using the software of Microsoft Office Excel 2007. Pairwise comparisons of methane yields from *Miscanthus floridulus* with different pretreatments were carried out by *z*-test and the significant level was set as 0.05. Statistical analysis results showed methane yields from pretreated *Miscanthus floridulus* (except hot water pretreated *Miscanthus floridulus*, $p = .09 > .05$) were statistically significant higher than that of control ($p < .05$). However, methane yields between different pretreatments had no statistically significant differences (except H_2O_2 pretreatment and hot water pretreatment, $p = .02 < .05$). Among these 5 pretreatment methods, H_2O_2 pretreatment obtained highest cumulative methane yield. However, H_2O_2 pretreatment needs considerable amounts of H_2O_2 and NaOH to maintain the pretreatment condition. Alkali pretreatment and acid pretreatment also need large amount of alkali or acid to proceed the pretreatment process. In addition, after alkali pretreatment, acid pretreatment and H_2O_2 pretreatment, additional chemicals (acid or alkali) also need to adjust the pH of the AD system. In comparison, microaerobic pretreatment also led to higher methane yield from *Miscanthus floridulus* (18.7% higher than that of control) and microaerobic pretreatment just needs limited oxygen or air, which is more economical and environmental friendly.

Modified Gomperz equation fitting results were shown in Table 3. All the R^2 were greater than 98.6%, which means the methane data could be well explained by the fitting results. Among these five pretreatment methods, maximum daily methane yield of 18.3 ml/g $VS_{substrate}$ per day

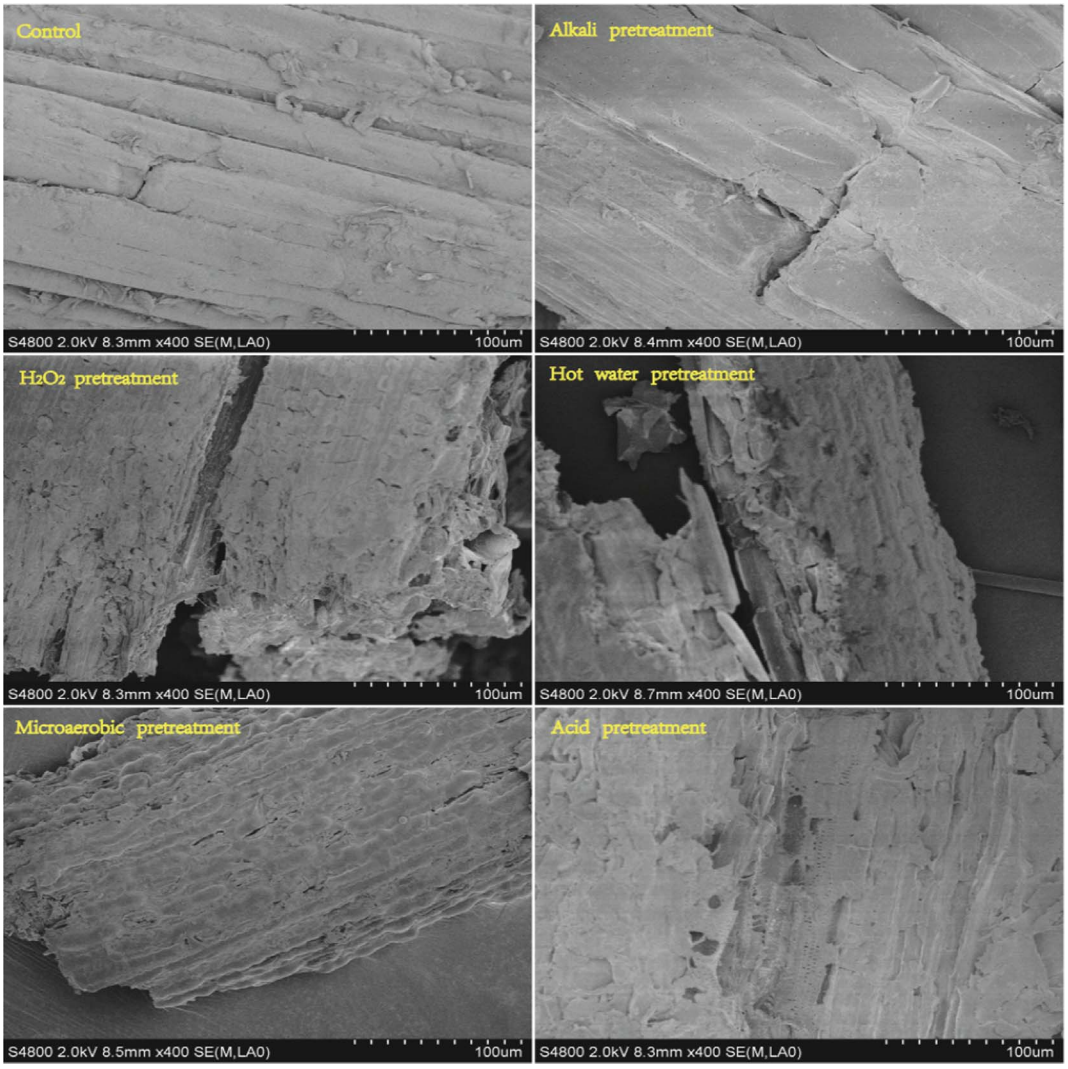


Fig. 2. SEM images of untreated and pretreated *Miscanthus floridulus* at 400×.

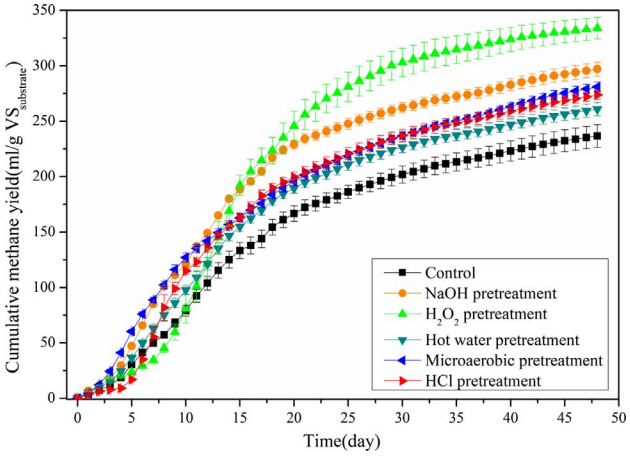


Fig. 3. Cumulative methane yield from *Miscanthus floridulus* during AD.

was obtained from H_2O_2 pretreated *Miscanthus floridulus*. Lag-phase time represents the time needed during batch AD until obvious methane production. As for this data, microaerobic pretreatment showed the best performance. Lag-phase time of microaerobic pretreatment group was 0, which means even no lag during the AD of microaerobic pretreated *Miscanthus floridulus*. After NaOH pretreatment, hot water pretreatment and microaerobic pretreatment lag-phase time decreased for 0.6, 0.4 and 2.2 days shorter than that of control, respectively, which means these pretreatment methods could accelerated the start-up of AD. However, after H_2O_2 pretreatment and HCl pretreatment, lag-phase time was prolonged, which could attribute to system acidified during the early AD stage of H_2O_2 pretreated and HCl pretreated *Miscanthus floridulus* (shown as Fig. 4).

3.3. The SCOD concentration and pH value variations during the AD of *Miscanthus floridulus*

The SCOD concentration and pH value variations during the AD of *Miscanthus floridulus* were shown in Fig. 4. The parameter of SCOD represents the solution extent of substrate, which is the dynamic equilibrium between the solution of substrate into SCOD and the consumption of SCOD to produce methane/carbon dioxide. During the initial stage of AD, the SCOD concentration of the fermentation system increased quite quickly. For pretreated groups the SCOD concentration reached maximum at the 6th day of AD. Relatively, the SCOD concentration of untreated group reached maximum at the 9th day of AD. Statistical analysis of the SCOD concentrations during the AD of *Miscanthus floridulus* were conducted using the software of Microsoft Office Excel 2007. Pairwise comparisons of SCOD concentrations with different pretreatments were carried out by *t*-test at the two-sided significance level $p = .05$. During the whole AD process, SCOD

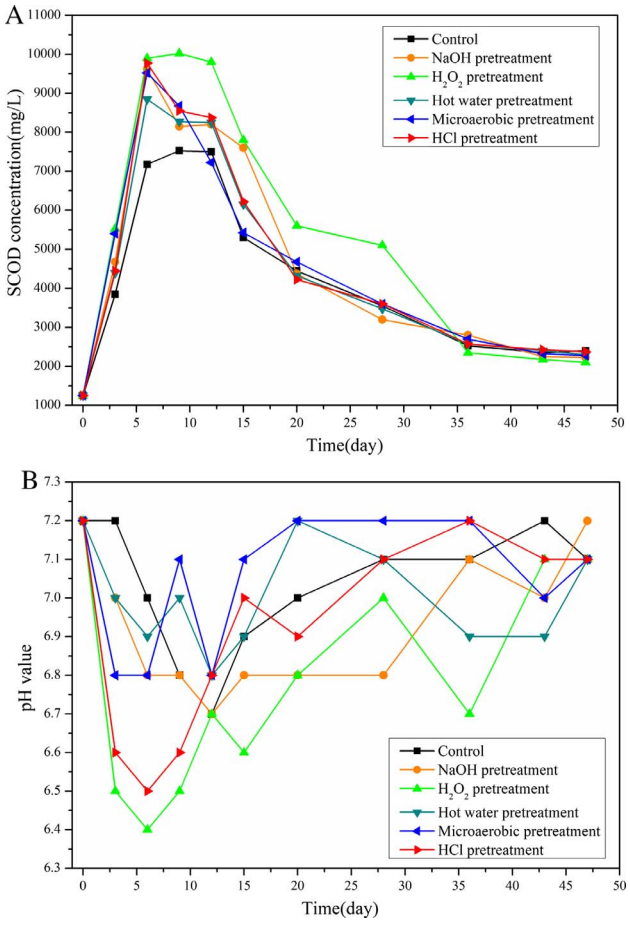


Fig. 4. SCOD concentration and pH value variations during the AD of *Miscanthus floridulus* (A: SCOD concentration B: pH value).

concentrations between different groups had no significantly differences ($p > .05$). However, during the initial AD stage (from 3rd day to 12th day), SCOD concentrations of pretreated groups were significant higher than that of un-pretreated group, which showed the solubility of substrate during the initial stage of AD improved obviously by pretreatment process, which might attribute to the destruction of dense structures during pretreatment.

The pH value is also a crucial parameter of AD. The optimal pH range for AD is 6.8–7.2 [47], a lower or higher pH value may lead inhibition to AD system. The pH values of NaOH pretreated, hot water pretreated, thermophilic microaerobic pretreated and un-pretreated groups were always among 6.8–7.2. Similarly, pairwise comparisons of pH with different pretreatments were carried out also by *t*-test at the two-sided significance level $p = .05$. Statistical analysis results showed pH during the whole AD process had no significant differences between different groups. However, pH of un-pretreated group during 3rd day to 9th day was significant higher than those of HCl and H_2O_2 pretreated groups. At the 6th day of AD, the pH value of HCl pretreated group dropped to 6.5 and recovered to more than 6.6, rapidly. Between the 3rd to 9th days of AD, the pH value of H_2O_2 pretreated group was lower than 6.6. During H_2O_2 pretreatment, the dense structure of substrate was destroyed more radical (as shown in Table 1 and Fig. 2), which led to higher hydrolysis rate in H_2O_2 pretreated group. These were also the reason for the longer lag-phase time of H_2O_2 pretreated group and lower biogas-production efficiency during this period (shown in Table 4 and Fig. S2).

Table 4
The *Modified Gomperz equation* fitted with methane production data.

Group	P (ml/g $VS_{substrate}$)	R_m (ml/g $VS_{substrate}$ per day)	λ (d^{-1})	R^2 (%)
Control	229.5 ± 1.7	10.0 ± 0.3	2.2 ± 0.4	99.5
NaOH pretreatment	284.9 ± 1.5	14.4 ± 0.5	1.6 ± 0.4	99.4
H_2O_2 pretreatment	327.4 ± 1.7	18.3 ± 0.6	5.2 ± 0.3	99.7
Hot water pretreatment	260.0 ± 1.7	12.0 ± 0.4	1.8 ± 0.4	99.5
Microaerobic pretreatment	271.6 ± 3.5	11.1 ± 1.1	0	98.6
HCl pretreatment	260.3 ± 2.6	13.2 ± 0.7	2.5 ± 0.5	98.9

Note: P is the total methane production potential (ml/g $VS_{substrate}$), R_m is the maximum methane production rate (ml/g $VS_{substrate}$ per day), λ is the lag-phase time (d) and t is the elapsed time (d).

3.4. Preliminary economic analysis of different pretreatment methods

The Preliminary economic analysis results were shown in Table 5. Due to the improved input caused by chemical/energy, the costs to produce per cubic meter biomethane of these four traditional pretreatments (alkali hydrogen peroxide pretreatment, NaOH pretreatment, HCl pretreatment and hot water pretreatment) were higher than that of control. Similar results were also widely reported in other studies. Study carried out by Passos et al. [18], who made the experimental and economic evaluation of thermochemical pretreatment during the AD of dairy cow manure. Even the methane potential of dairy cow manure could be significantly improved after thermal-alkali or thermal-acid pretreatment, the untreated group was more economically feasible. The economically feasible of pretreatment might be affected by pretreatment types and conditions applied and improved methane yield after pretreatment. As Ruffino Barbara et al. demonstrated if NaOH pretreatment was carried out at dose of 0.08 g NaOH/g TS, 60% increase of the methane yield could offset the cost of chemicals [48]. In order to improve the economical performance of chemical/thermal pretreatment methods, researches of chemical (e.g. NaOH, HCl) reuse technologies and energy recovery technologies during the pretreatment process are urgently needed. Among these five pretreatment methods, only the microaerobic pretreatment group obtained a lower cost than un-pretreated group to produce per cubic meter biomethane. Meanwhile, during the microaerobic pretreatment, there were no additional chemicals needs. Therefore, compared with other pretreatment methods, microaerobic pretreatment is more economical and environmental friendly.

4. Conclusions

The technical and preliminary economical analysis of five pretreatment methods was conducted in the present study. The AD efficiency of *Miscanthus* could be significantly improved by the pretreatment process. Different pretreatments have unequal effects on the structure of *Miscanthus*. Among the five pretreatment methods, alkali hydrogen peroxide pretreatment was most efficient in improving the methane yield, however the start-up performance of microaerobic pretreated *Miscanthus* was best. Preliminary economical analysis showed only microaerobic pretreatment among these five pretreatment methods was economically feasible. In order to improve the economical performance of chemical/thermal pretreatment methods, researches of chemical (e.g. NaOH, HCl) reuse technologies/energy recovery technologies during the pretreatment process are urgently needed. Based on the results obtained in current study, microaerobic pretreatment is an attractive pretreatment method which is both economically feasible and environmental friendly during the AD of cellulosic substrate (e.g. *Miscanthus*).

Table 5
Preliminary economic analysis of different pretreatment methods (the semi-continued data was designed based on batch experiment results and model fitting results).

Groups	HRT(day)	Daily Cost for substrate (U.S. $)	Daily cost during pretreatment process				Daily cost during start-up of AD		Total daily cost (U.S. $)	Daily methane yield (m^3)	Economic analysis (U.S. $/$m^3$ methane)
			Energy needed during pretreatment (kJ)	Chemical needed during pretreatment (kg)	Cost for heating during pretreatment (U.S.$)	Cost for chemical during pretreatment (U.S.$)	Energy needed during start-up/ running stage (kJ)	Cost for heating during start-up/ running stage (U.S.$)			
NaOH pretreatment	20	21.78	$3.696 * 10^5$	NaOH 48 HCl 38.4	11.55	51.11	$1.478 * 10^6$	46.19	130.63	222.71	0.587
H_2O_2 pretreatment	18	24.20	$4.107 * 10^5$	H_2O_2 177.78 NaOH 4.48 HCl 3.24	12.83	97.28	$1.643 * 10^6$	51.34	185.65	278.19	0.667
Hot water pretreatment	22	19.80	$2.352 * 10^6$	–	73.50	–	–	–	93.30	177.79	0.525
Microaerobic pretreatment	24	18.15	$1.344 * 10^6$	–	42.00	–	$1.96 * 10^5$	6.12	66.27	175.53	0.378
HCl pretreatment	20	21.78	$2.735 * 10^6$	HCl 29.2 NaOH 24	85.47	32.81	–	–	140.06	205.24	0.682
Control	23	18.94	–	–	–	–	$1.607 * 10^6$	50.22	69.16	154.29	0.448

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.enconman.2018.01.014.

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